

Lecture 1: Introduction to Low-Resource Task Adaptation

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*Presenter
Video*

*Available in English,
Cantonese and Mandarin*



JOHN DONNE

(English Poet, 1572-1631)

>> *“No man is an island, entire of itself; every man is a piece of the continent, a part of the main.”*
- *Devotions upon Emergent Occasion (1624)*

COURSE PATTERN

- Short Lectures
 - Each lecture is 20-60 minutes long and focuses on one specific topic
 - The lectures cover the main concepts and key knowledge points of the tutorial
- After-Class Exercises
 - Critical reading and reflection on related research
 - Surveying recent advances in related research directions
 - Hands-on coding projects
- Communication
 - GitHub Issues for public discussions, including questions about the tutorials and coding projects hosted under the GitHub Organization.
 - Email is reserved for personal questions

CONTENT

- What Will You Learn from This Tutorial?
 - Understand the basic definition of large language models (LLMs) ★☆☆^[1]
 - Learn the general process of developing and deploying LLMs ★☆☆
 - Recognize the low-resource challenges of applying LLMs to real-world tasks ★☆☆
 - Get a bird's-eye view of low-resource task adaptation ★★☆☆

OUTLINE

- What Are Large Language Models (LLMs)?
 - History and Recent Applications
 - Evolution of LLMs
 - Deployment Pipeline for Downstream Tasks
- Are LLMs Perfect for Every Task?
 - Scarcity of Task-Specific Data
 - Three Key “How” Questions
- Low-Resource Task Adaptation
 - Problem Formulation and Framework
 - Taxonomy of Related Work
- Summary ● Bibliography for Further Study
- Exercises ● Index
- Acknowledgements

[1] The stars indicate the difficulty level of each learning objective.

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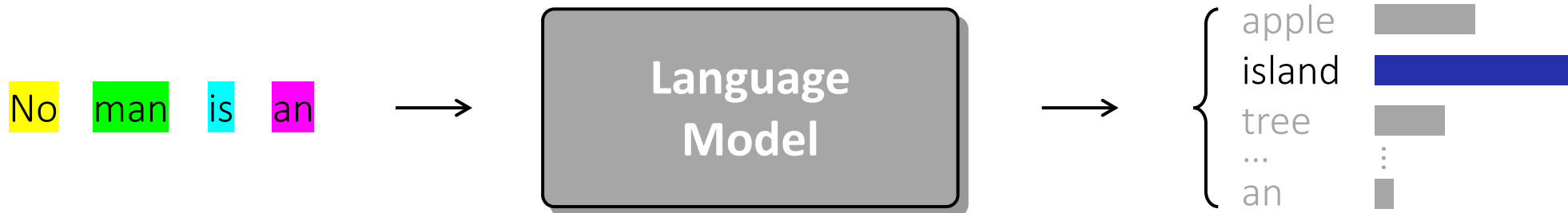
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What Are Language Models (LMs)?

- Everything starts from language modeling.
 - How can a computer understand languages and generate human-like outputs?
 - Traditional computational view: statistical models
 - Human-inspired view: neural models



Statistical Models

N-Gram LM	P
Unigram: apple	0.5
Bigram: an apple	0.6
Trigram: is an apple	0.8
...	

Counting word co-occurrences in **discrete** space

Neural Models

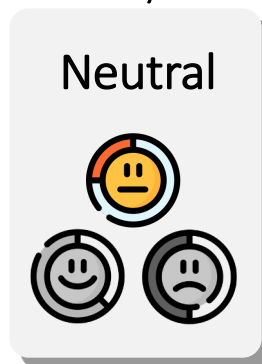
Learning representations and predicting words in **continuous** space

What Are Language Models (LMs)?

- Everything starts from language modeling.
 - How can a computer understand languages and generate human-like outputs?
 - Traditional computational view: statistical models
 - Human-inspired view: neural models
 - With the understanding and generation capabilities, LMs can do more NLP tasks.
 - Sentiment analysis
 - Machine translation
 - Information retrieval
 - Even tasks beyond plain text (e.g., speech recognition, handwriting recognition)

Besides Chinese and Western influences, Hong Kong's culture is enriched by its indigenous communities and ethnic minority communities with roots in South and Southeast Asia.

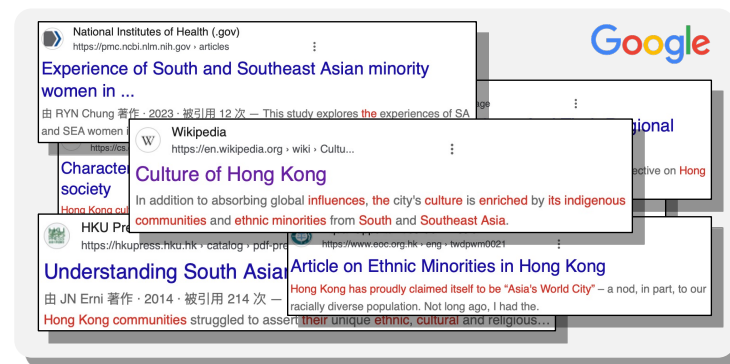
↓ Sentiment Analysis



↓ Machine Translation

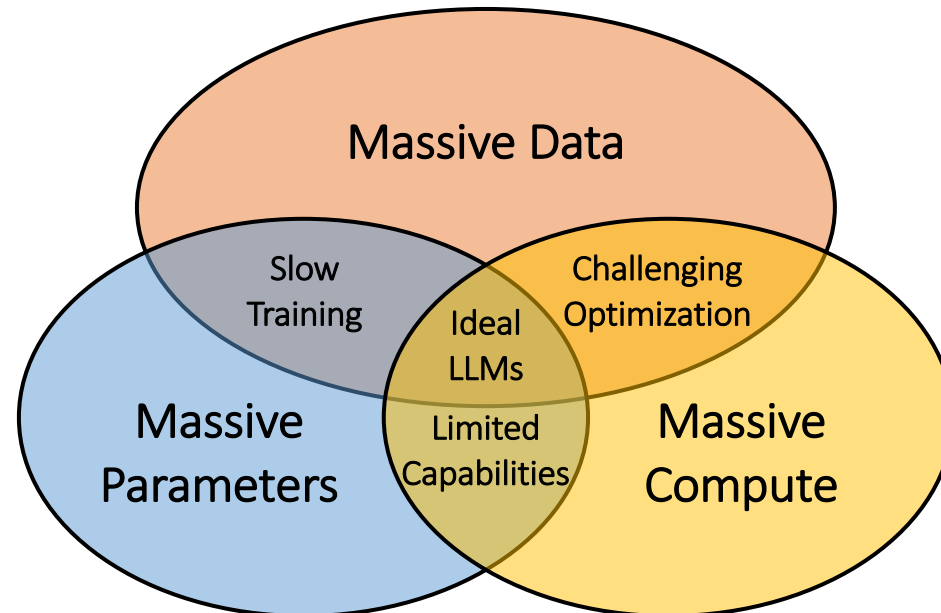
除了中國和西方的影響外，香港的文化還因其本土社區和源自南亞和東南亞的少數民族社區而更加豐富多彩。

↓ Information Retrieval



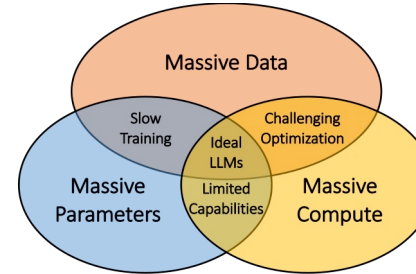
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- An explosion of language capability - **large** language models (LLMs)
 - **Large** amounts of data
 - **Large** numbers of neural parameters
 - **Large**-scale computing resources

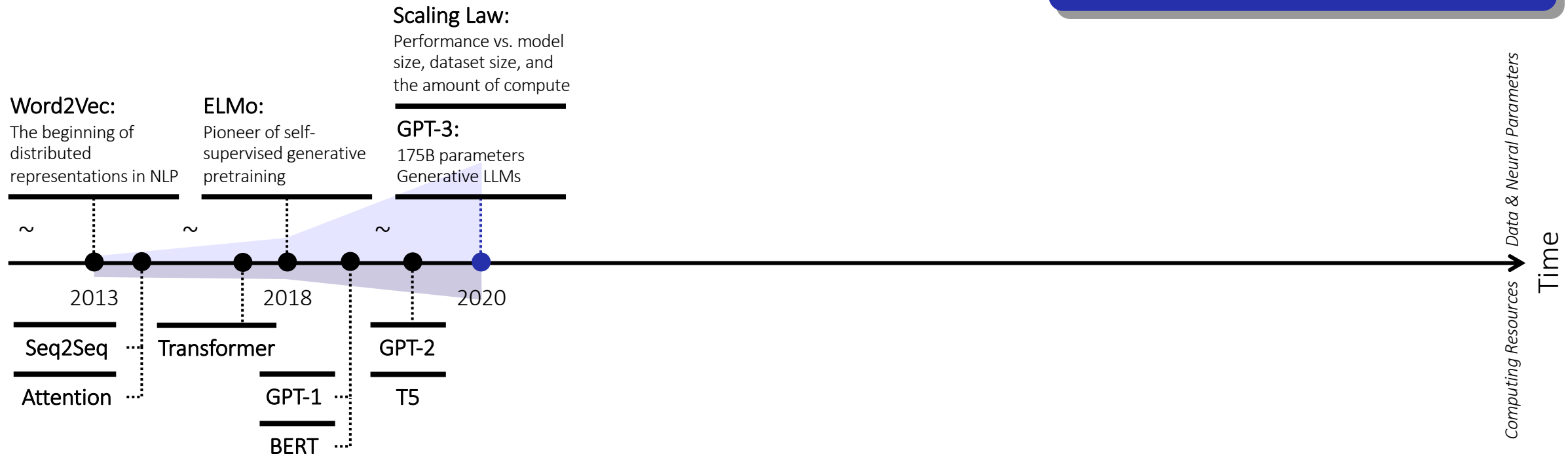


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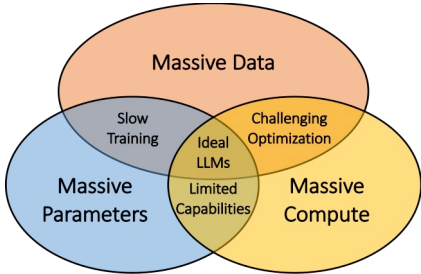


2020: Breakthrough of LLMs

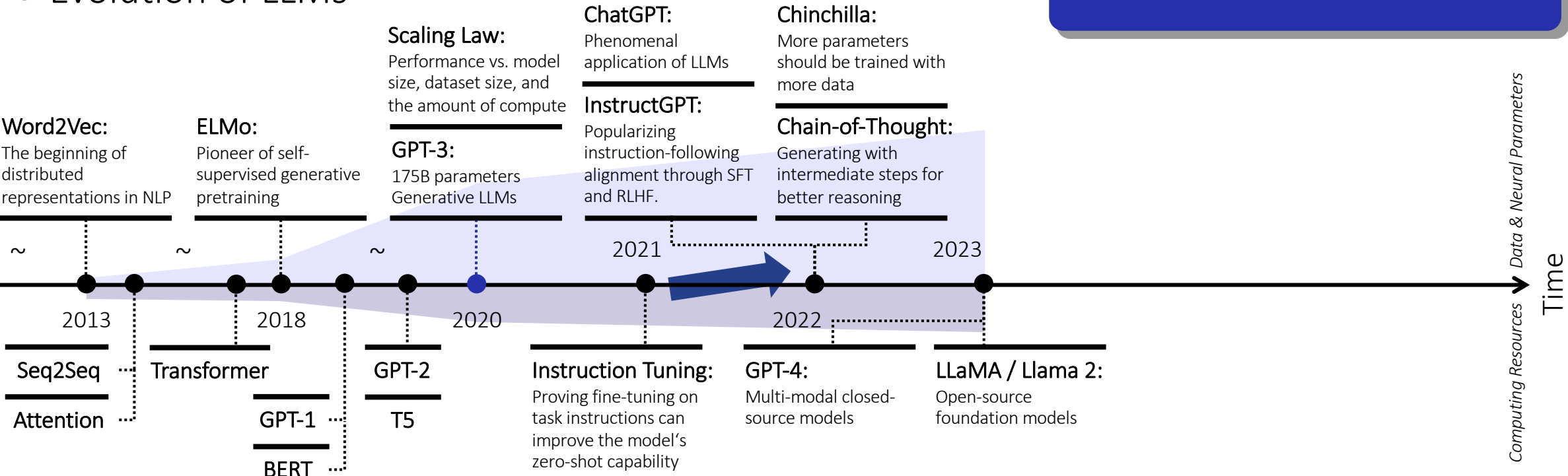


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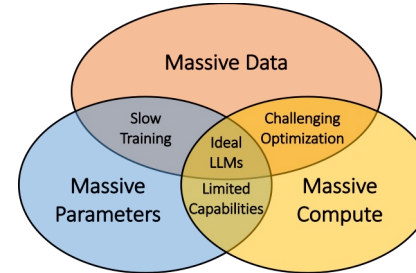
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2020-2023: Scaling up with more data and more parameters



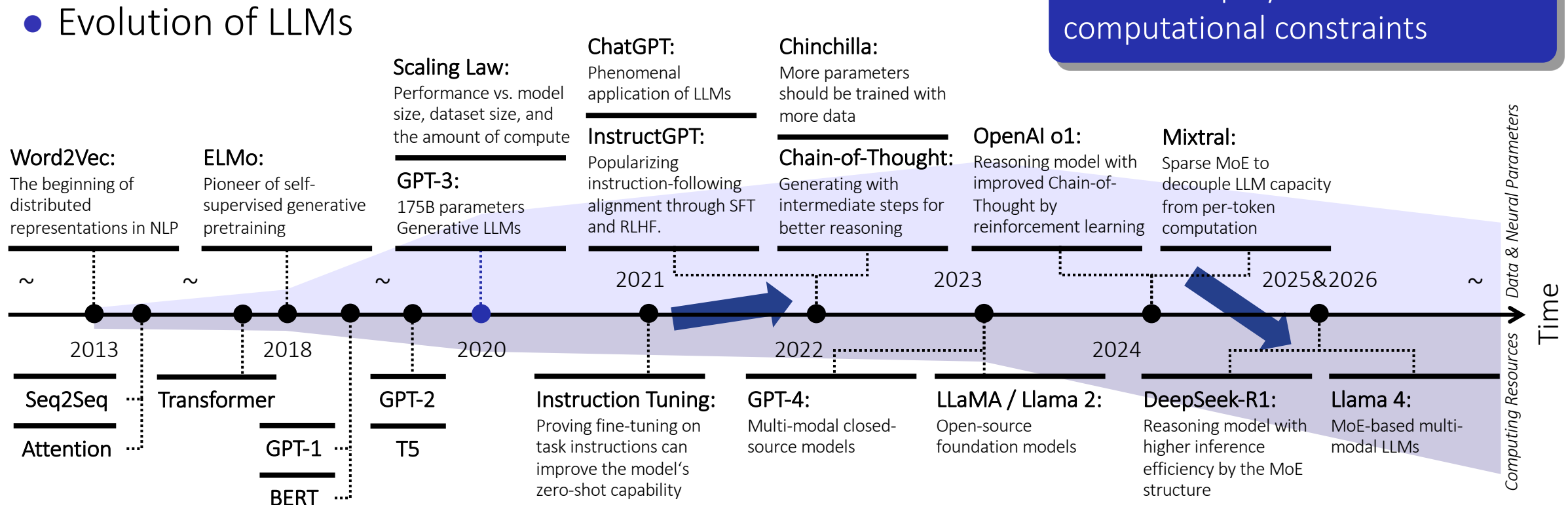
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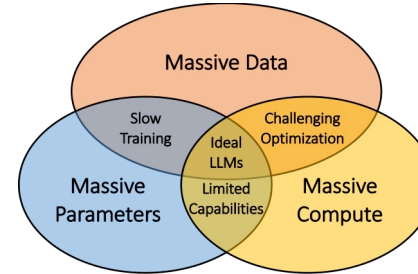


2020: Breakthrough of LLMs
2020-2023: Scaling up with more data and more parameters
2023-present: Shifting toward efficient deployment under computational constraints



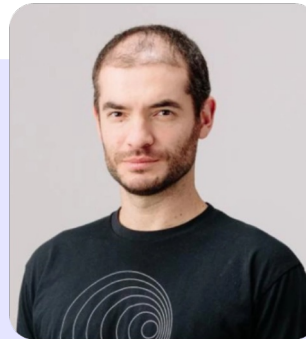
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“Data is not growing, because we have but one Internet. You could even go as far as to say that data is fossil fuel of AI. It was created some how, and now we use it, and we’ve achieved peak data and there’ll be no more [1].”

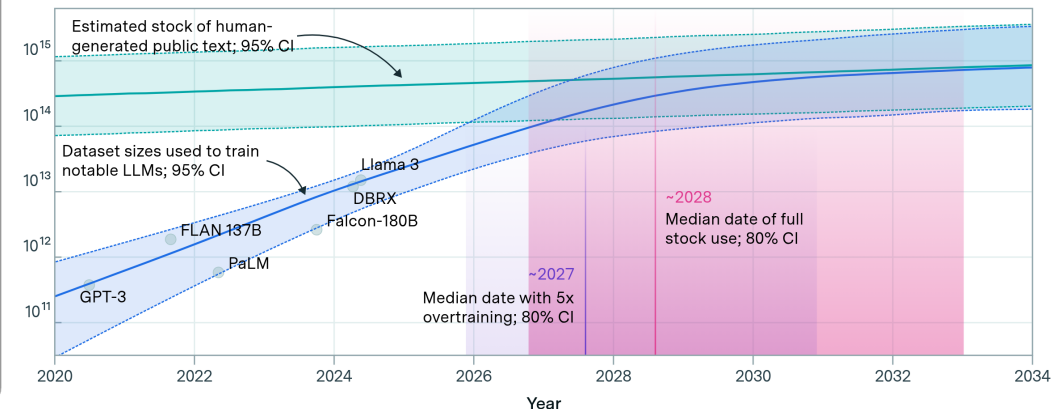


Ilya Efimovich
Sutskever
(Cofounder and Chief
Scientist of OpenAI)

Projections of the stock of public text and data usage

EPOCH AI [2]

Effective stock (number of tokens)

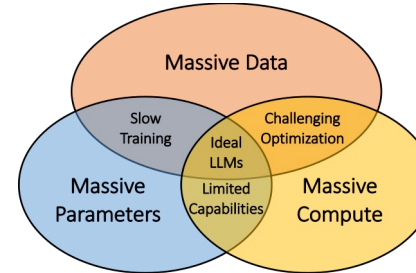


[1] <https://www.youtube.com/watch?v=1yvBqasHLZs&t>

[2] <https://epoch.ai/publications/will-we-run-out-of-data-limits-of-llm-scaling-based-on-human-generated-data>

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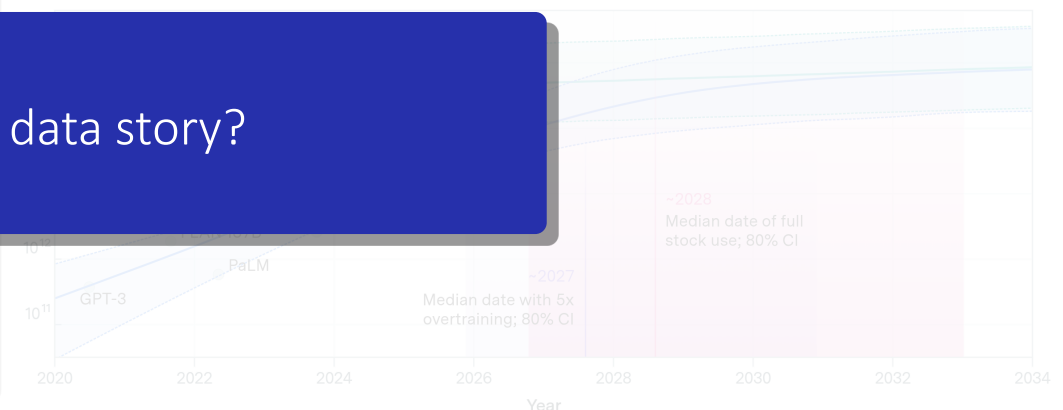
“Data is not growing, because we have but one Internet. You could even go as far as to say that it is fossil fuel of AI. It will run out some how, and not long, and we’ve achieved peak data and there’ll be no more [1].”

So, this is **the end** of the data story?

Sutskever
(Chief Scientist of OpenAI)

Projections of the stock of public text and data usage

Effective stock (number of tokens)



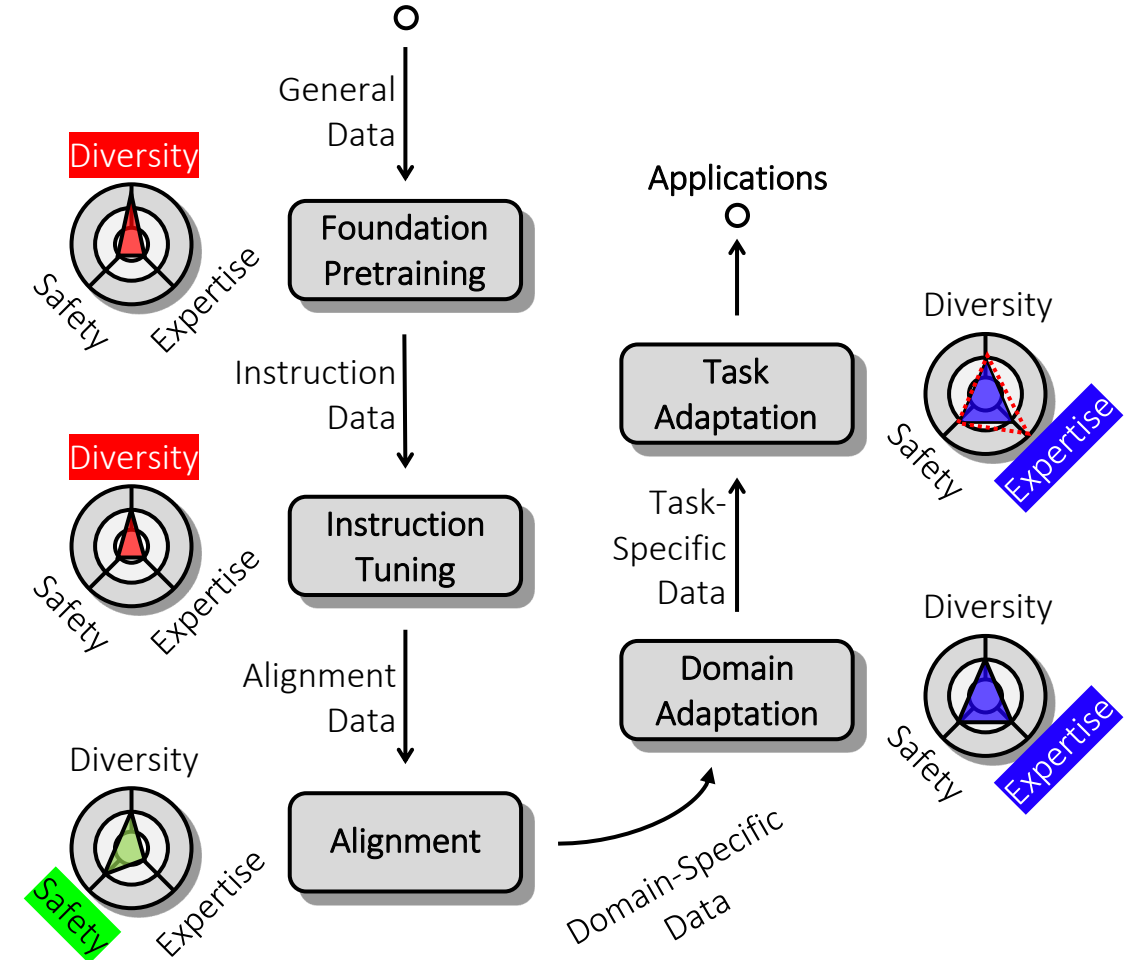
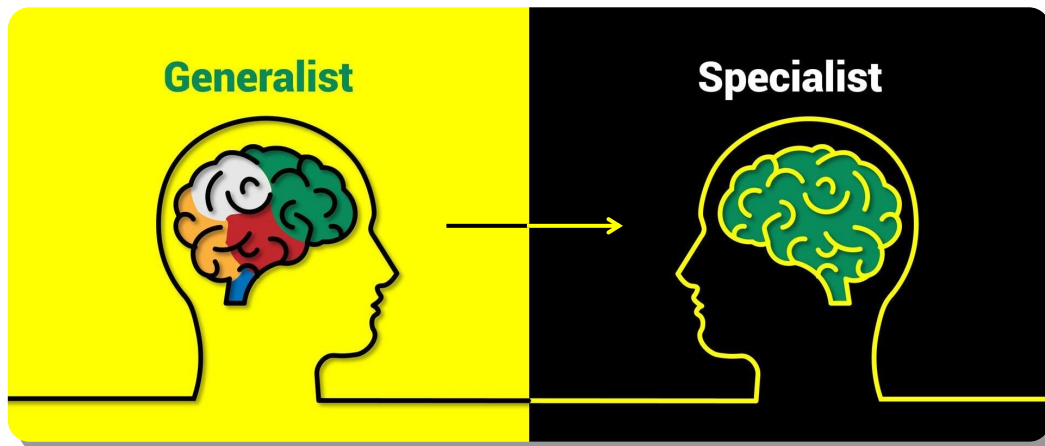
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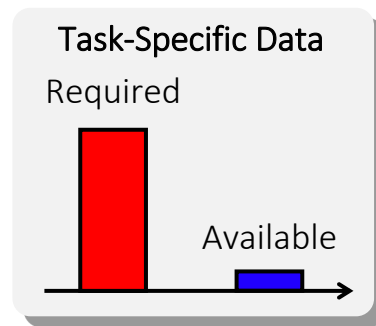
- Let's take a look at a typical deployment pipeline first!
 - Foundation pretraining
 - Instruction tuning
 - Alignment
 - Model adaptation
 - Domain adaptation
 - Task adaptation



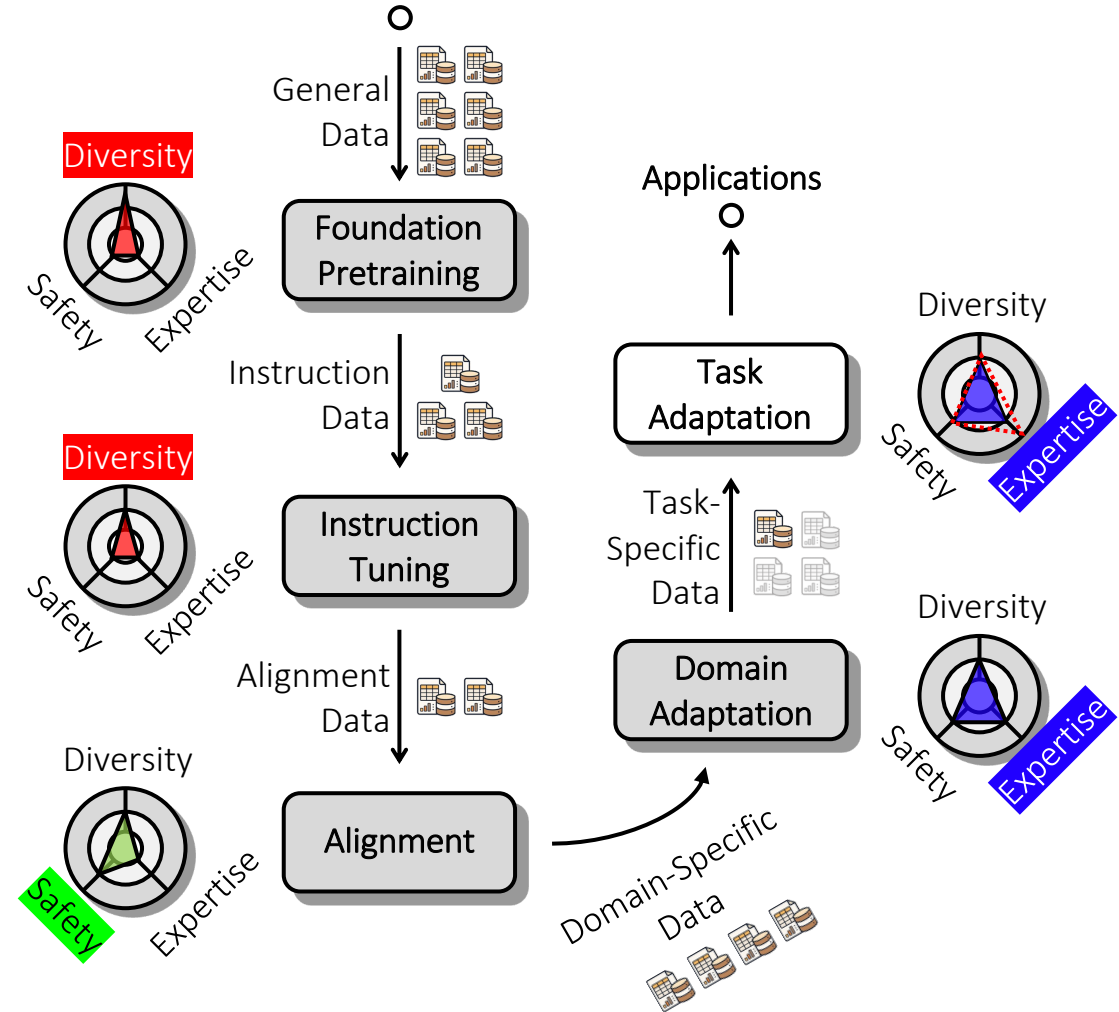
Are LLMs Perfect for Every Task? - Not really!

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 - Task adaptation
 - In-domain
 - Task-specific
 - Labeled
 - High-quality

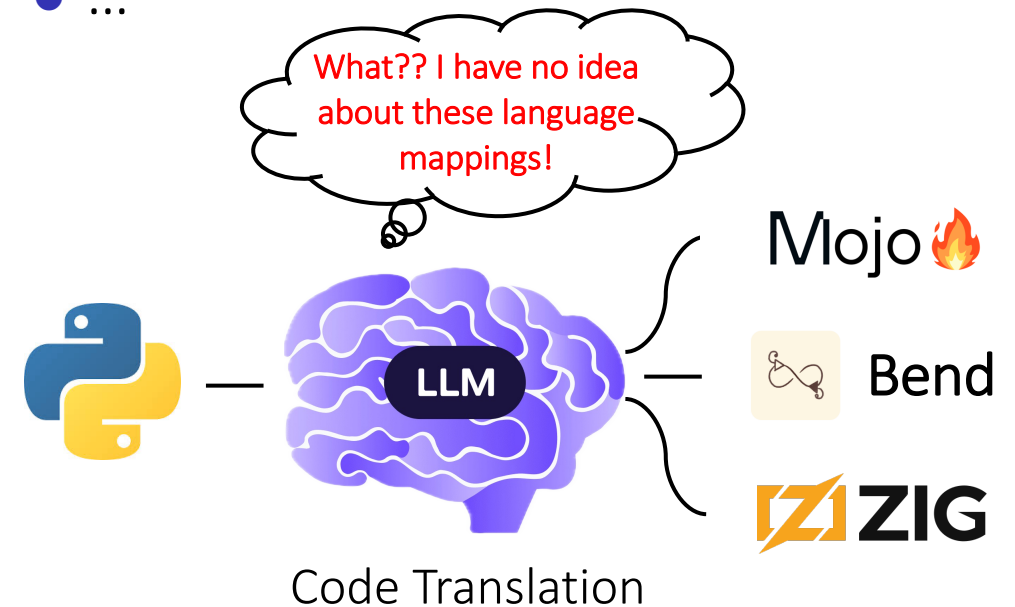
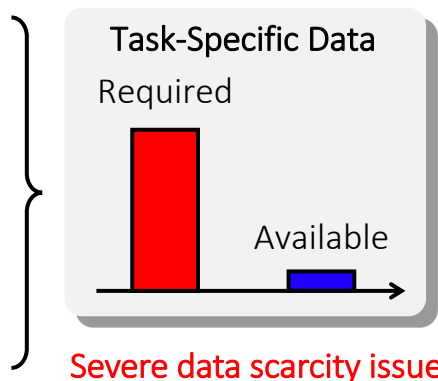


Severe data scarcity issue



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 - In-domain
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 - High-quality
- The data scarcity issue becomes **more serious** in rapid explosion of new tasks.
 - Emerging programming languages
 - Agent tasks
 - ...



Are LLMs Perfect for Every Task? - Not really!

- Let's take a look at a typical deployment pipeline first

- Foundation pretraining

- The data scarcity issue becomes **more**

- **Low-resource Task Island**

- Scarce task-specific labeled data.
- Limited benchmarks and supervision.
- Underserved needs and domains.

- **Mainland of NLP Progress**

- Abundant data and benchmarks.
- Diverse tasks and general capabilities.
- External knowledge and tools.

No Task Is an Island:
Low-Resource Task Adaptation in LLMs
Connecting low-resource tasks to broader NLP progress through data, training, and inference.

• High-quality



Severe data scarcity issue



Code Translation

ZIG



Are LLMs Perfect for Every Task? - Not really!

- Three key “**how**” questions

- 1 **How** can we obtain and leverage high-quality supervision?
- 2 **How** can we perform effective model updating under limited labeled data?
- 3 **How** can we improve reliability at inference time for low-resource LLMs?

Low-Resource Task Adaptation

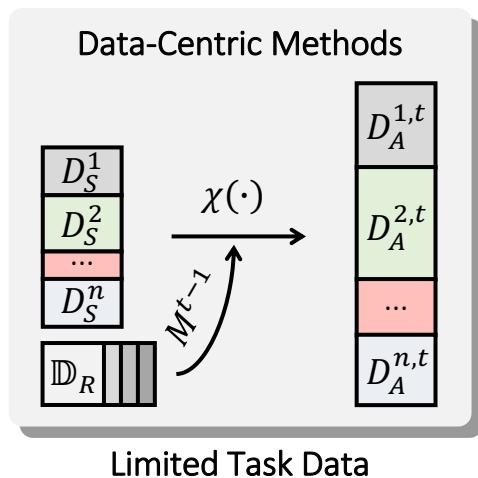
- Three key “**how**” questions 1 2 3
- Problem formulation and framework

- Data-centric methods

- $\{D_A^{1,t}, \dots, D_A^{n,t}\} = \chi(\{D_S^1, \dots, D_S^n\}; M^{t-1}, \mathbb{D}_R)$

- Number of training tasks: n
- Post-training step: t
- Seed labeled datasets: $\{D_S^1, \dots, D_S^n\}$
- Datasets applied at the t_{th} post-training step: $\{D_A^{1,t}, \dots, D_A^{n,t}\}$
- Model parameters from the last training step: M^{t-1}
- Set of auxiliary unlabeled data resources: $\mathbb{D}_R$
- Data-centric procedure: $\chi(\cdot)$

Large Language Model M^0



- 1 **How** can we obtain and leverage high-quality supervision?

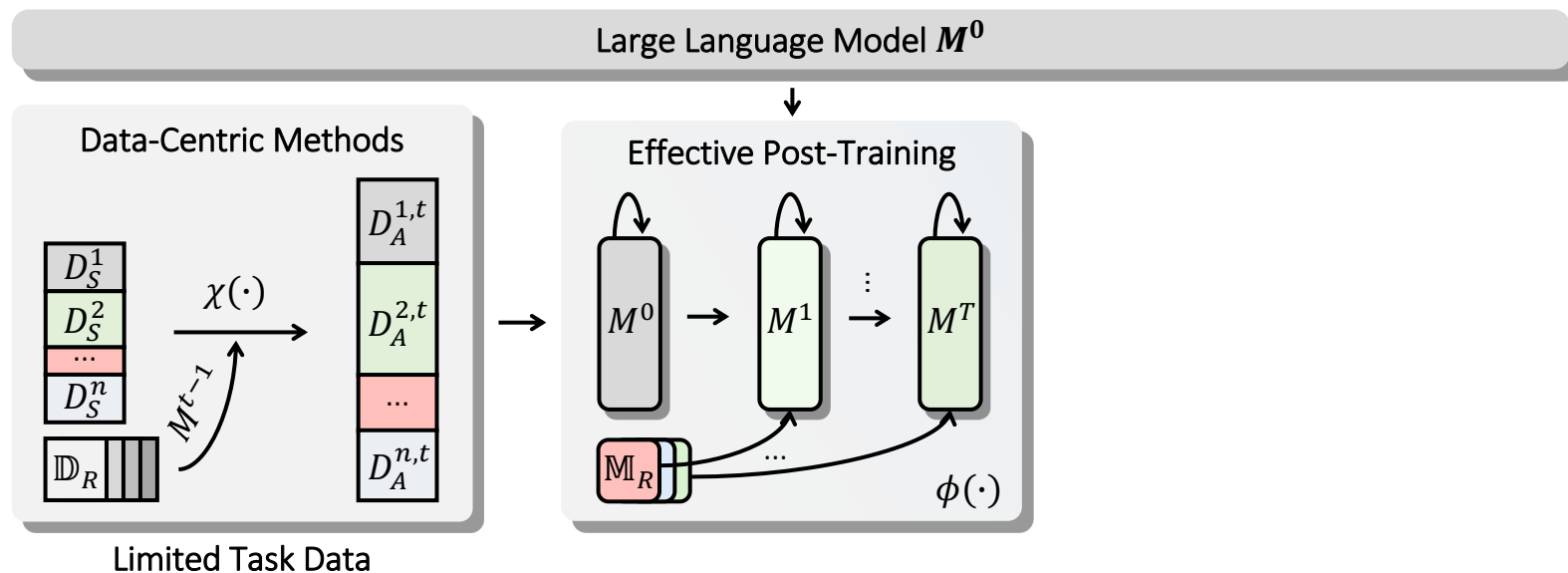
Low-Resource Task Adaptation

- Three key “**how**” questions 1 2 3
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- Effective post-training

- $M^t = \phi(\{D_A^{1,t}, \dots, D_A^{n,t}\}, M^{t-1}; \{M^0, \dots, M^{t-2}\}, \mathbb{M}_R)$

- Number of training tasks: n
- Post-training step: t
- Model parameters from training step t and $t - 1$: M^t and M^{t-1}
- Datasets applied at the t_{th} training step: $\{D_A^{1,t}, \dots, D_A^{n,t}\}$
- Earlier model checkpoints: $\{M^0, \dots, M^{t-2}\}$
- Set of auxiliary models: $\mathbb{M}_R$
- Post-training function: $\phi(\cdot)$



- 2 **How** can we perform effective model updating under limited labeled data?

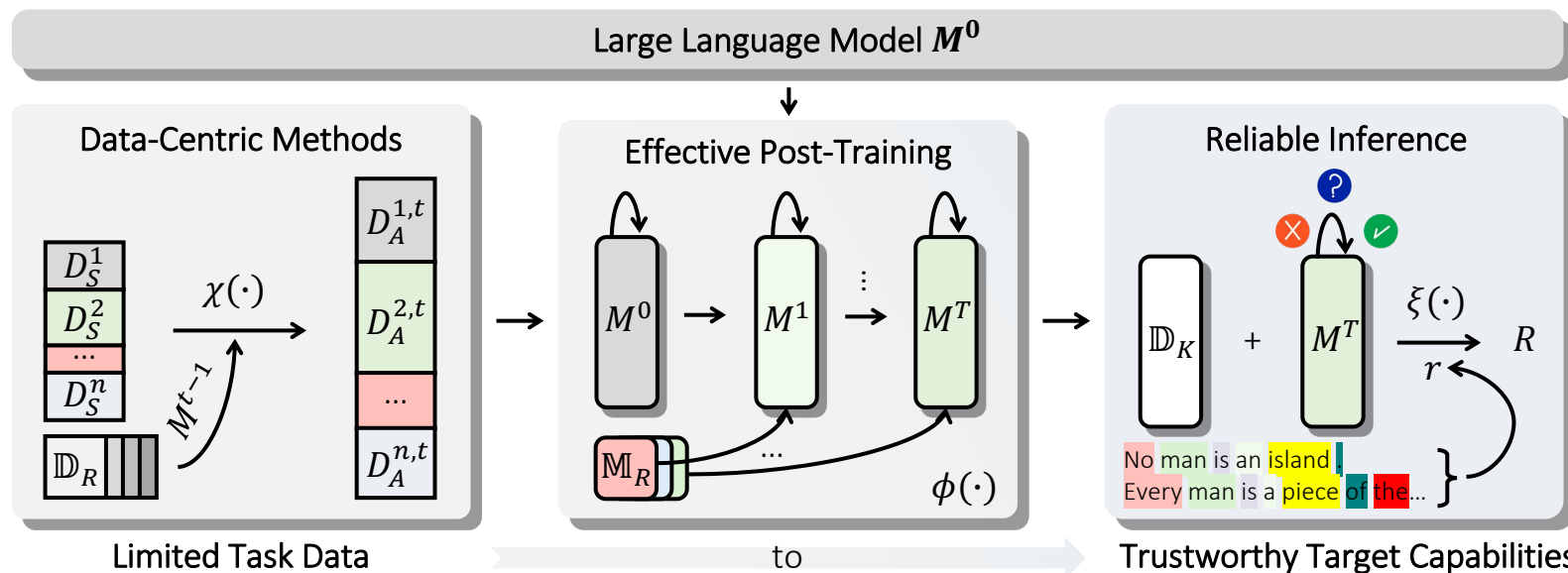
Low-Resource Task Adaptation

- Three key “**how**” questions ① ② ③
- Problem formulation and framework

- Data-centric methods
- Effective post-training
- Reliable inference

$$R = \xi(Q, M^T, r; \mathbb{D}_K)$$

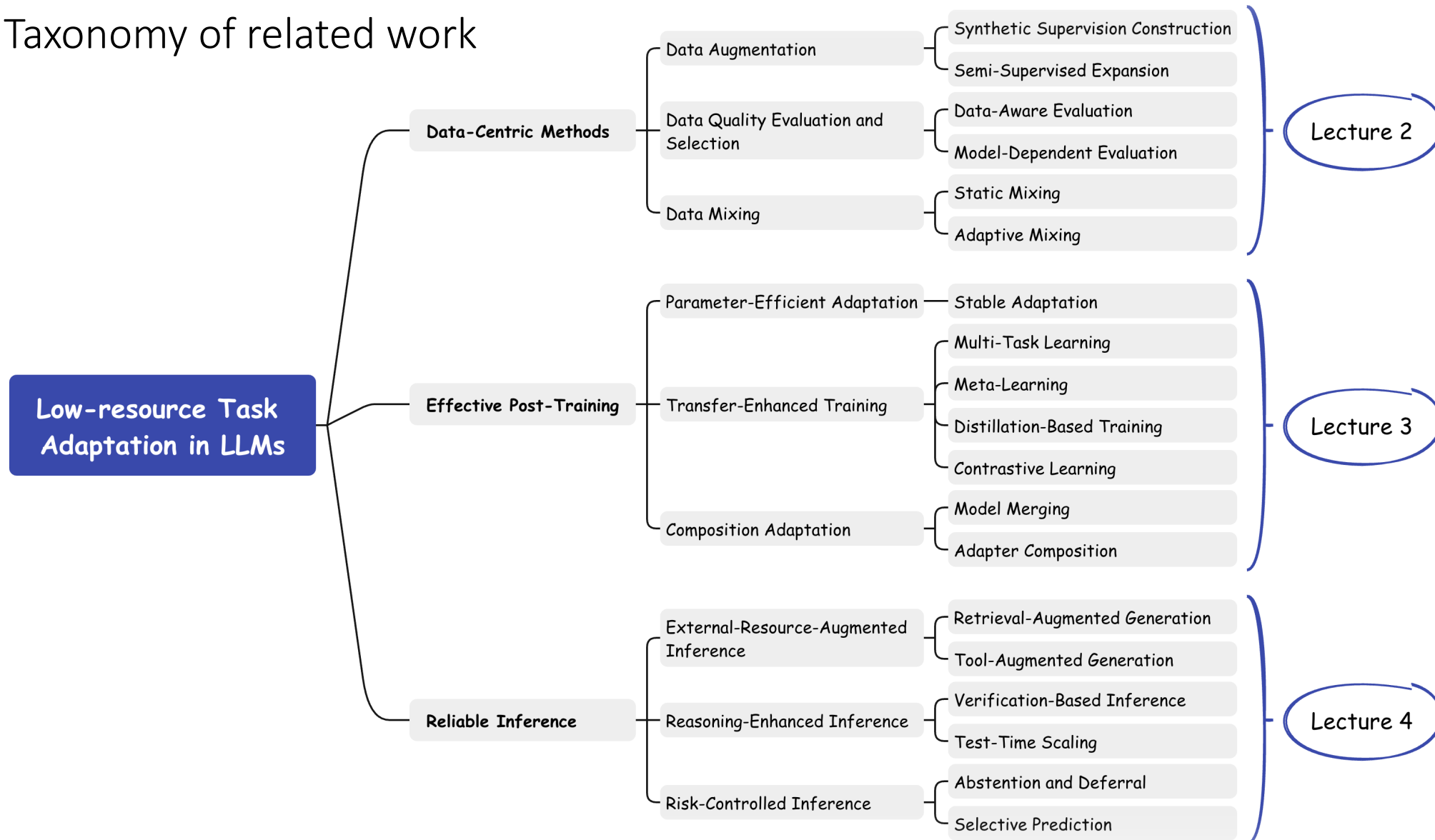
- Number of post-training steps: T
- Adapted LLM: M^T
- Query: Q
- Response: R
- Inference strategy: r
- External knowledge or tool base: $\mathbb{D}_K$
- Inference process: $\xi(\cdot)$



- ③ **How** can we improve reliability at inference time for low-resource LLMs?

Low-Resource Task Adaptation

- Taxonomy of related work



Now, Let's Go Back to The Question



So, this is **the end** of the data story?

Definitely not!

- Internet-scale data is being increasingly exhausted for LLM training.
- As a result, leading LLMs are expected to acquire broad world knowledge more comprehensively.
- However, **LLM capabilities remain highly uneven across tasks**. For low-resource tasks, especially newly emerging ones, **achieving effective task adaptation with limited data** still matters for LLM research and development.

Summary

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Wrap up of this lecture!

- Understand the basic definition of large language models (LLMs). ★☆☆
- Learn the general process of developing and deploying LLMs. ★☆☆
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- Get a bird's-eye view of low-resource task adaptation. ★★★



Exercises

Task 1

In what kinds of tasks do LLMs have greater potential than humans? Give some examples. What skills are still difficult for LLMs to master in the near future?

Task 2

Which low-resource tasks are urgent and socially important? Can current LLM systems address them effectively? Survey related open-source projects for one or two such tasks.

Task 3

Based on your background, what machine learning approaches can help alleviate data scarcity in low-resource LLM adaptation?



Please feel free to submit your exercise results and feedback to our GitHub repository (<https://github.com/liyingqiao2012/Low-Resource-Task-Adaptation-in-LLMs/issues>).

exercise

discussion

Bibliography for Further Study

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Hong Kong
Sign Language^[1]



香港手語協會
Hong Kong Sign Language Association

English
Thank you! $\longleftrightarrow$



An Example of Low-Resource Task:
English-Hong Kong Sign Language Machine Translation

*Presenter
Video*

[1] The sign-language image is from the Hong Kong Sign Language Association (<https://www.hksla.org.hk/everybody/014.jpg>).